



International
Online
Case
Competition



BRIGHTWATER ETHIOPIA

CLEAN WATER FROM BROKEN WELL
PRESENTED BY WAI CHEW TEAM

SUSTAINABLE BUSINESS CASE IOCC 2023

EXECUTIVE SUMMARY



BrightWater’s current aspiration to be sustainably with affordable and clean water using our approach to reach the goals.

Goal

To became more sustainably, replacing broken wells with solar power solution and shifting drinking water to plastic bottle

Key Question

“Without supporting from NGOs, how can BrightWater continue to provide more clean water to local community?”

Recommendations

EXPAND SOLAR POWER WELL

Increasing more clean water sales by expanding to 11 solar power wells

SHIFT TO PLASTIC BOTTLE

Leaning cost of product by partner with Addis Mercato

E-Mail Message

Marketing e-mail messages, by targeting baby boomer donors

Impacts

90%

Water accessibility to population

150%

Increase in water pump efficiency to reservior

35%

Compound annual growth rate (CAGR) during 2021-2025

ETHIOPIA'S OVERVIEW

Current situation that affects water shortage and unclean in Dilla, Ethiopia.

GEOGRAPHY OF ETHIOPIA



PROBLEMS

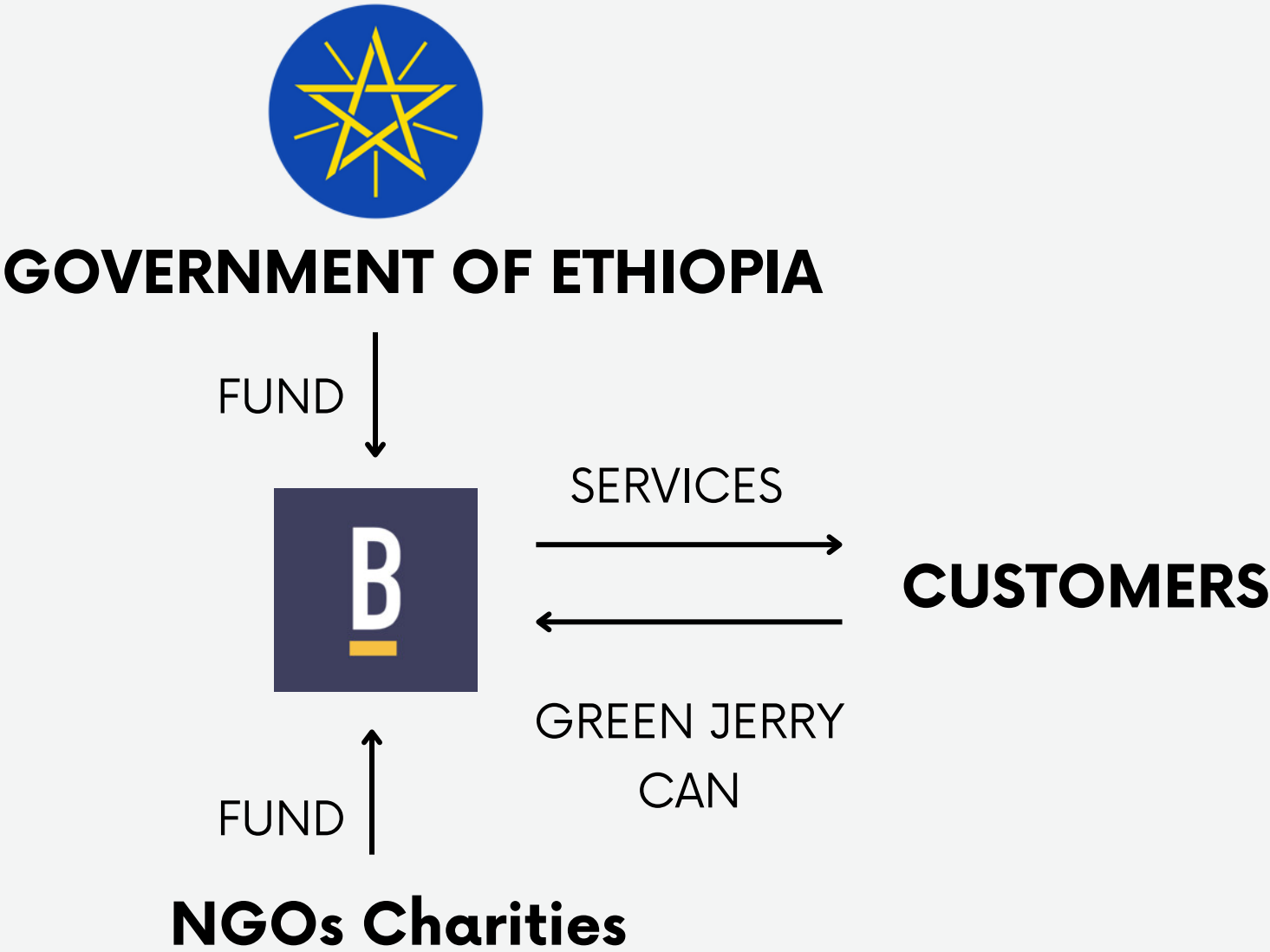
- Current water only served population coverage of 61.9%
- High cost of production with Brightwater alone
- Lack of government support and subsidies
- Current inflation rates in Ethiopia which affect construction cost, yet labors are lack of skills.
- The impossible of accessibility of Dilla's downtown as the civil war and conflicts are still occurring.
- Low potential of water capacity due to power outage from high cost of power generators and leakage of pipes that are connect to eleven of boreholes

BRIGHTWATER'S OVERVIEW

Current BrightWater's situation and missions aim to increase more affordable and safe drinking water.



BUSINESS MODEL



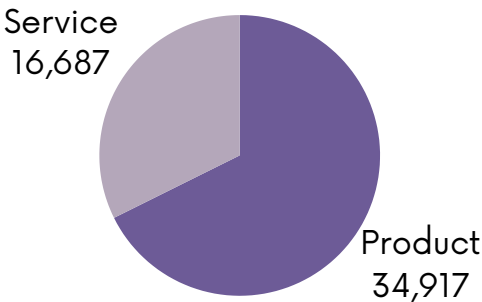
VISION

To reduce incidence of water-borne disease by increasing access to affordable, safe drinking water.

MISSION

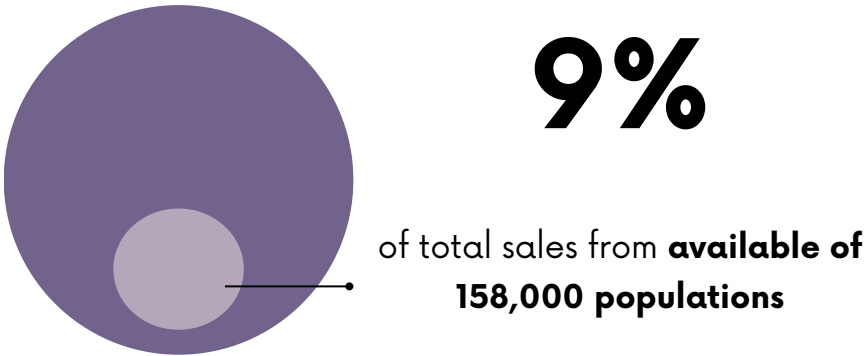
Focusing on feasibility, sustainability, scalability and social impact.

MAIN REVENUE



67.7%
of total revenues are from **clean water**

MARKET OPPORTUNITIES



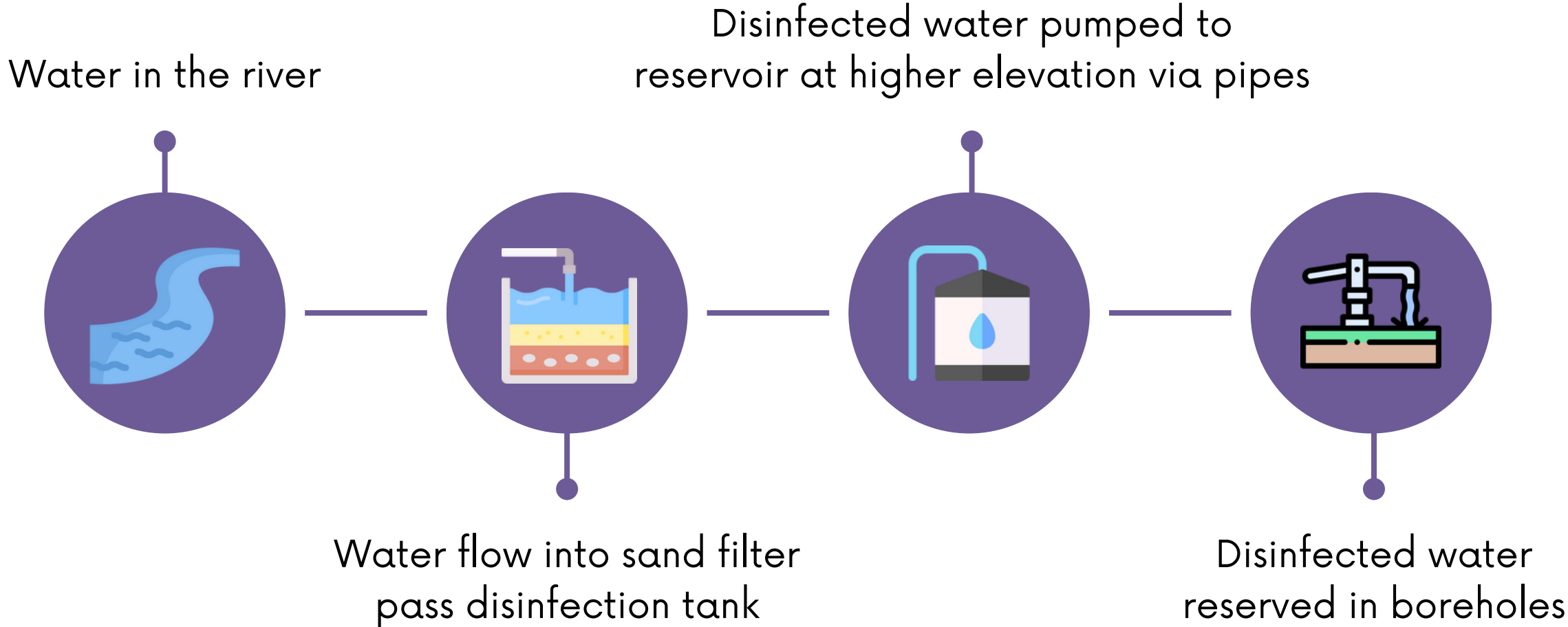
SITUATION ANALYSIS

WAI CHEW'S SOLUTION

IMPLEMENTATION

BRIGHTWATER'S WATER SUPPLY PROCESS

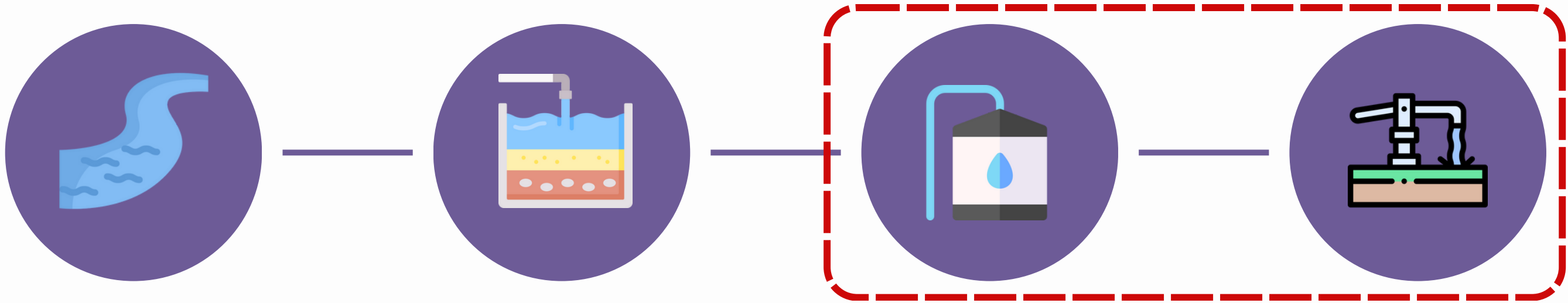
The processes of Brightwater's water supply from the beginning to the people's hand.



BRIGHTWATER'S BARRIERS OF WATER SUPPLY



Water supply's pain points are segmented into two key barriers to address.



LACK OF SYSTEMATIC VERIFICATION

CONSUMERS SAFETY
They don't know if there's coalugulant inside the cholorine barrel and needed to be replaced.

RESULT
People consumes contaminants unconsciously lead to mortality

POOR PUMP POWER AND INSPECTION

INEFFICIENTCY POWER
The reservoir doesn't have a power generator as a result of the pump stops working sometimes when they are power outage which occurs daily and high frequency. The pump power limits at capacity of 200 cubic meters.

WATER PRODUCTION INEFFICIENT
The water lost in leakage of pipes is estimated to be more than 22% before reaching the boreholes.

RESULT
The water production is inefficient and wasted large amount of water.

BRIGHTWATER'S BARRIERS OF WATER SUPPLY

KEY QUESTION

What are opportunities for BrightWater to grow the business?

RESULT

People consumes contaminants unconsciously lead to mortality

WATER PRODUCTION INEFFICIENT

The water lost in leakage of pipes is estimated to be more than 22% before reaching the boreholes.

RESULT

The water production is inefficient and wasted large amount of water.

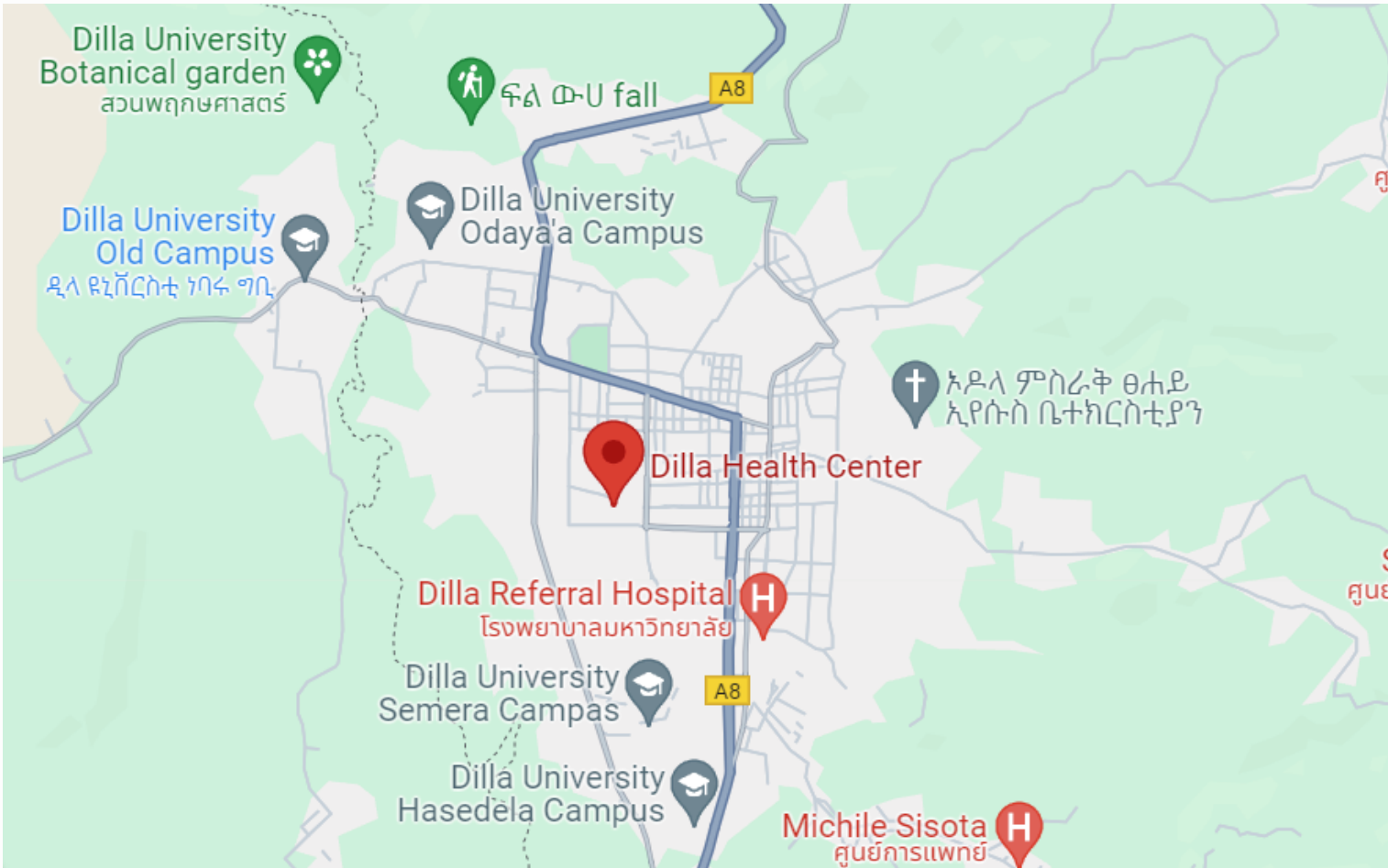
EMERGENCY WATER STORAGE: OUR SOLUTION



SYSTEM MODEL



SUCCESS PARTNER



ULTIATE GOAL: As BrightWater builds RWH System at Dilla’s Health Center and focuses on repairing training and management strategy. We expected it will be harvested enough for Dilla’s populations to storage water to use when there’s an emergency seasonally.

ONLINE FUNDRAISING IN 2022



To fundraising more equities without supporting from NGOs, BrightWater should target invidual donors American.

\$499.33 BILLIONS WERE DONATED TO CHARITY BY AMERICAN

MILLENNIAL DONORS WORLDWIDE

PREFER TO GIVE	
Online (credit/debit card)	55%
Cash	14%
Bank/Wire Transfer	11%
PayPal	9%

MOST INSPIRED TO GIVE BY	
Social Media	39%
Email	23%
Website	20%
Direct Mail/Post	6%

GEN X DONORS WORLDWIDE

PREFER TO GIVE	
Online (credit/debit card)	55%
Bank/Wire Transfer	12%
Cash	10%
PayPal	10%

MOST INSPIRED TO GIVE BY	
Social Media	33%
Email	26%
Website	19%
Direct Mail/Post	9%

BABY BOOMER DONORS WORLDWIDE

PREFER TO GIVE	
Online (credit/debit card)	54%
Direct Mail/Post	19%
Bank/Wire Transfer	10%
PayPal	8%

MOST INSPIRED TO GIVE BY	
Email	33%
Social Media	19%
Direct Mail/Post	18%
Website	16%

TARGET DONOR

KEY TAKEAWAYS

BrightWater can raise the maximum funds by targeting individuals in aged “Baby Boomers” by email marketing, here is a example of email message interface,

Dear Michael,

As the holiday season approaches, a time for reflection and giving, I wanted to share a cause close to my heart: the urgent need for clean water in Dilla, Ethiopia.

Imagine this: Every day, thousands of people in Dilla walk miles to collect water from contaminated sources, putting their health and livelihoods at risk. Children miss school due to waterborne illnesses, and women spend hours fetching water, leaving them little time for other pursuits. This is a reality we can change. With the help of [Charity Name], a reputable organization dedicated to providing clean water and sanitation solutions, we can build a sustainable water supply system in Dilla.

This would mean:

- Safe drinking water: Villagers would have access to clean, readily available water, reducing the spread of waterborne diseases.
- Improved health: Children wouldn't miss school due to illness, and families would be healthier overall.

Your contribution, no matter how big or small, can make a life-changing difference.

Here are some ways you can help:

- Donate directly: Every dollar donated goes towards building the water supply system in Dilla. Click here to donate securely: [Donation Link]
- Spread the word: Share this email with your friends and family, and encourage them to join the cause.

Michael, this holiday season, let's give the gift of life by bringing clean water to Dilla. Together, we can make a lasting impact on the lives of thousands of people.

Thank you for your time and consideration.

Sincerely,

BrightWater

[DONATE NOW](#)

Source: doublethedonation.com

SITUATION ANALYSIS

WAI CHEW’S SOLUTION

WAI CHEW’S SOLUTION

SHIFT TO REUSABLE PLASTIC BOTTLES



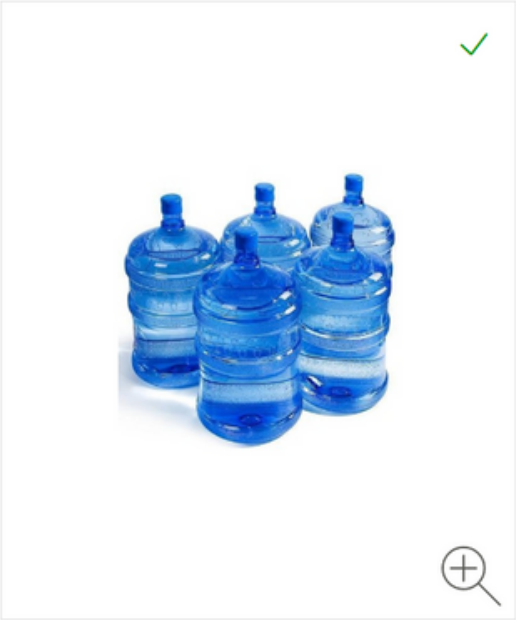
In additions of expanding to open 11 broken wells, BrightWater needs to lean cost of supply to gain maximum profits.

NEW SUPPLIER



← Previous

Next product →



PURE DRINKING WATER - 20-LITER (5.2-GALLON)

In stock (913 items available)

★★★★★

- Pure Drinking Water - including the container
- 20-Liter (5.2-Gallon)
- Delivered to your home

ETB575.00

1

ADD TO CART

 Buy Now
The safer, easier way to pay

Source: addismercato.com

COMPARAISON OF JERRY CAN AND PASTIC BOTTLE COST

GREEN JERRY CAN 20 LITRES
old water container

ETB 396
per container

PLASIC BOTTLE 20 LITRES
new water container

ETB 115
per container

KEY TAKEAWAYS

In order BrightWater to gain more profits, they have to lean cost of container. A shift to plasic bottle, BrightWater can save their cost by 70%. With power of negotiation, in the future, they can request for lower cost of supply.

SITUATION ANALYSIS

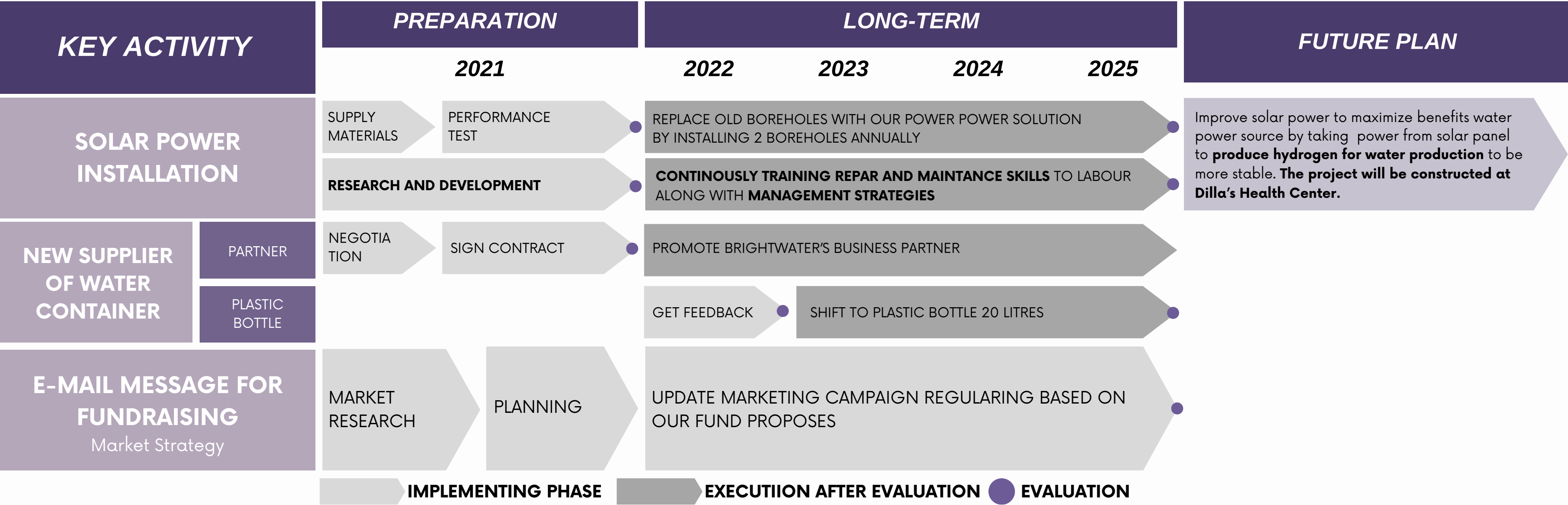
WAI CHEW’S SOLUTION

IMPLEMENTATION

IMPACT JUSTIFICATION

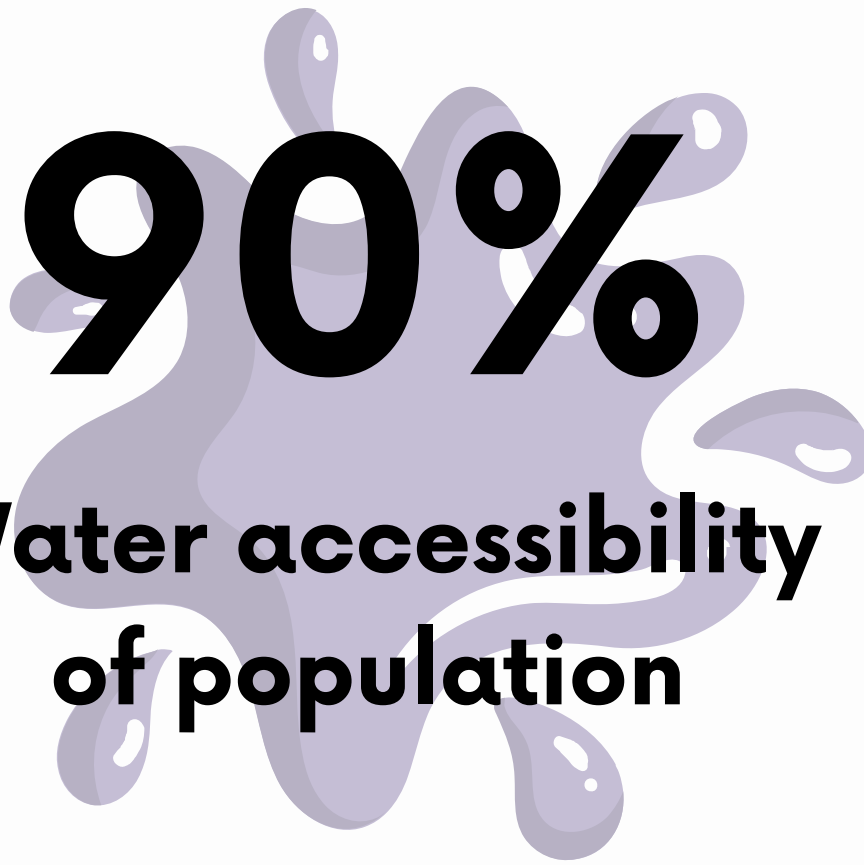
IMPLEMENTATION PLAN

BrightWater will reach their goals within preparation period and will continue to develop sustainably with solid foundation.



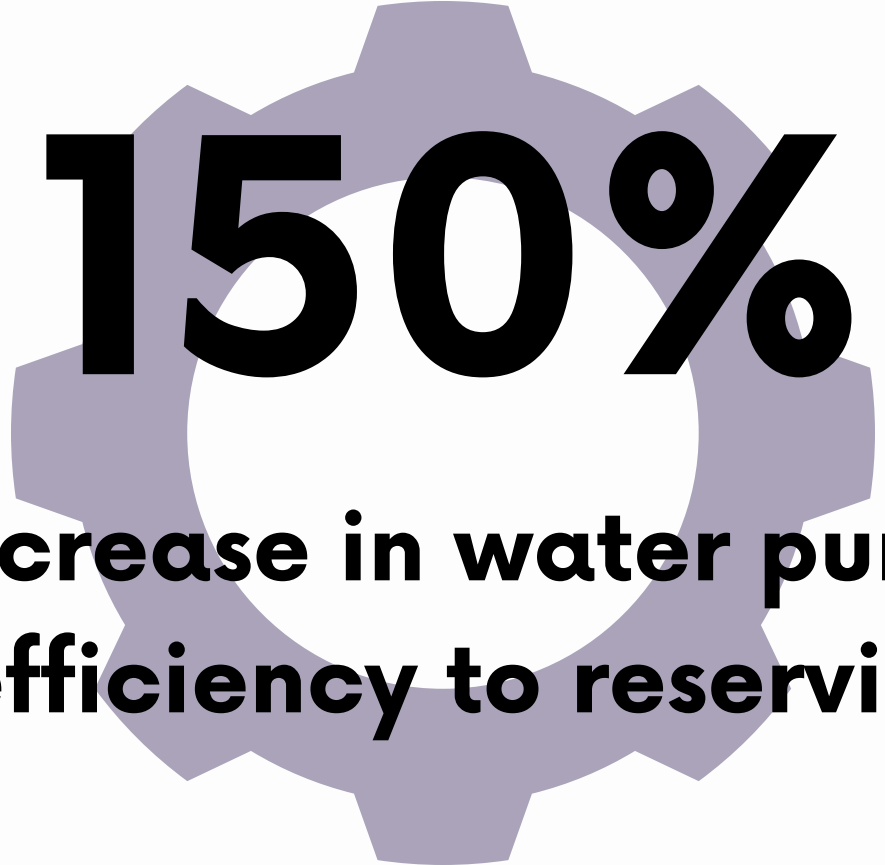
KEY PERFORMANCE INDICATORS

To fundraising more equities without supporting from NGOs, BrightWater should target invidual donors American.

A light purple water splash graphic with several droplets, serving as a background for the 90% KPI.

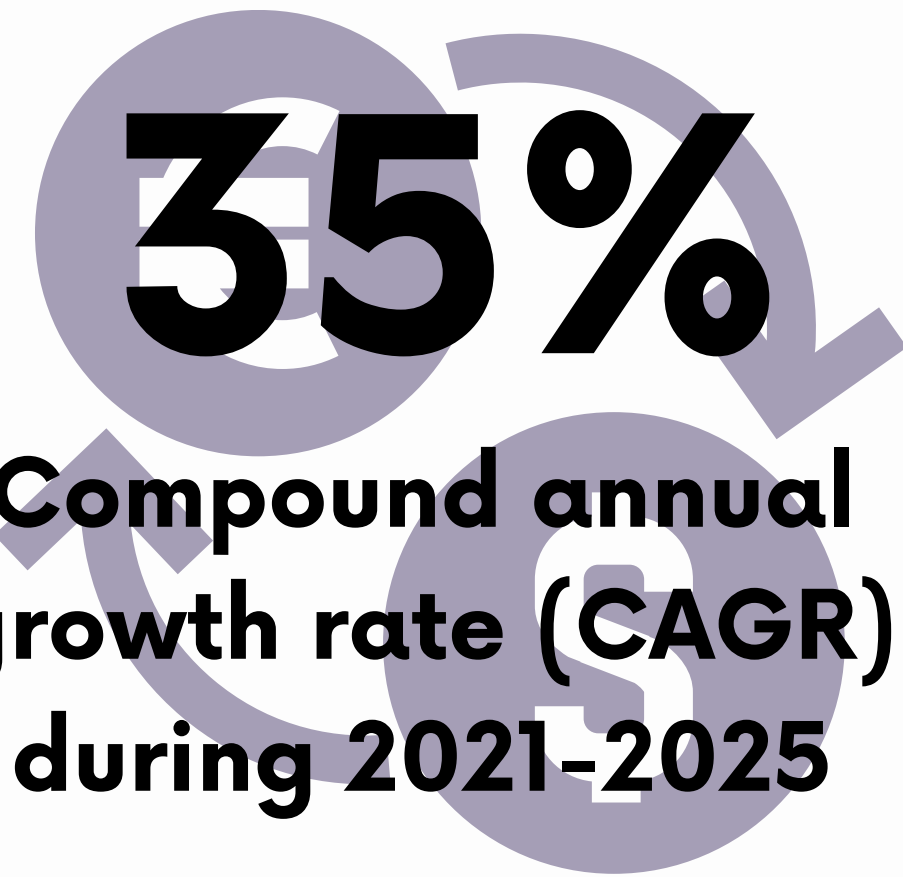
90%

**Water accessibility
of population**

A light purple gear graphic, serving as a background for the 150% KPI.

150%

**Increase in water pump
efficiency to reservior**

A light purple circular arrow graphic, serving as a background for the 35% KPI.

35%

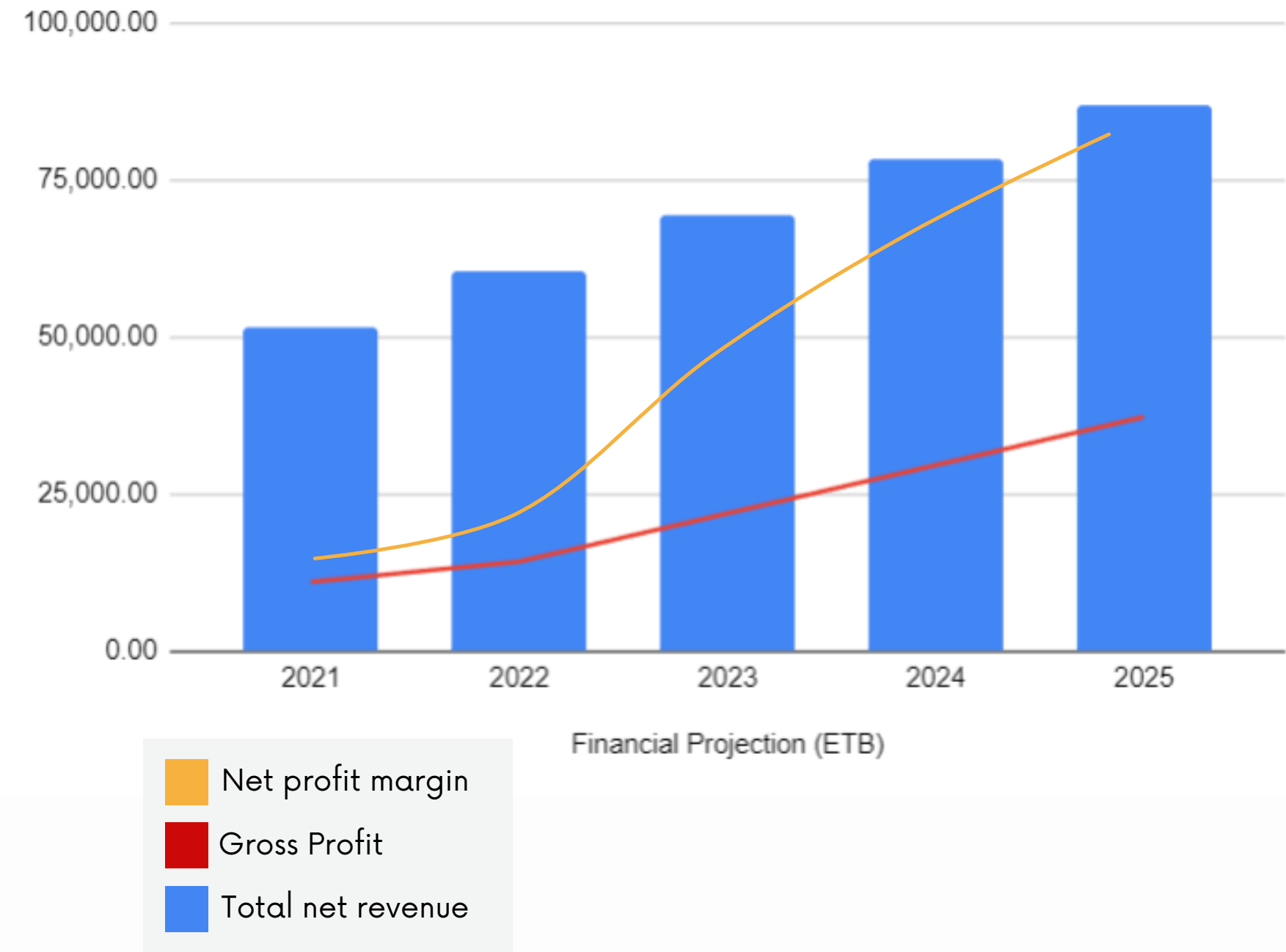
**Compound annual
growth rate (CAGR)
during 2021-2025**

FINANCIAL PROJECTION

To fundraising more equities without supporting from NGOs, BrightWater should target invidual donors American.



COMPARAISON OF BENEFIT AND COST



IMPACT

66.67%

Growth of
Customer rate

1.7 years

Pay back Period

51.70%

ROI in next the 4 years

14%

CAGR in next the 4 years

4-YEAR PLAN

**8 SOLAR
POWER WELLS
AND
CONSTANT SERVICE
OPERATION**

SITUATION ANALYSIS

WAI CHEW'S SOLUTION

IMPLEMENTATION



WAI CHEW TEAM
THANK YOU FOR YOUR TIME

FINANCIAL PROJECTION

BrightWater	Present	Forecast			
Financial Projection (ETB)	2021	2022	2023	2024	2025
Number of jerry cans sold	13,739.00	17,236.00	20,733.00	24,230.00	27,727.00
Clean water revenue	34,917.00	43,804.46	52,691.91	61,579.37	70,466.82
Shower service revenue	16,687.00	16,687.00	16,687.00	16,687.00	16,687.00
Total net revenue	51,604.00	60,491.46	69,378.91	78,266.37	87,153.82
Cost of goods sold	1,960.00	1,744.81	2,098.82	2,452.82	2,806.82
Gross profit	49,644.00	58,746.64	67,280.10	75,813.55	84,347.00
Email Marketing Expenses		10%	10%	10%	10%
		5,874.66	6,728.01	7,581.35	8,434.70
Operating expenses	38,550.00	44,424.66	45,278.01	46,131.35	46,984.70
Earnings before interest and taxes	11,094.00	14,321.98	22,002.09	29,682.19	37,362.30
Interest expense	0.00	0.00	0.00	0.00	0.00
Income taxes	0.00	0.00	0.00	0.00	0.00
Net earnings	11,094.00	14,321.98	22,002.09	29,682.19	37,362.30

ASSUMPTION AND CALCULATION

Initial Investment per well (ETB)	
Storage Construction	669.69
Contingency	2.26
Prepaid labour expenses	149.61
Solar panels	872.15
Water pump, plumber	778.71
Storage 5,000 litres (2)	700.83
Total initial investment	3,173.25

Comparison cost of water container	
Green jerry can 20L.	396
Plastic bottle 20L.	115
Reduction of cost of package(%)	70.96%

Assumption cost of services (shower and handwashing) based on income statement, 2021			
Number of solar powered wells	2		
Cost of installation per well	3173.25		
Total cost of installation	6346.5		
Operating Expenses	38,550		
Cost of service	32,204		

Assump growth rate of customers per one solar powered being installed						
	2020	2021	2022	2023	2024	2025
Number of solar powered wells	1	3	5	7	9	11
Number of solds	10,242.00	13,739.00	17,236.00	20,733.00	24,230.00	27,727.00

New cost of water container					
Number of solds	13739	17236	20733	24230	27727
Cost of supply per one sold	0.14	0.10	0.10	0.10	0.10
Cost of goods sold	1,960.00	1,744.81	2,098.82	2,452.82	2,806.82

ASSUMPTION AND CALCULATION



Assumption operating expenses	2022	2023	2024	2025
Number of solar powered wells	2	2	2	2
Cost of service (constant)	32,204	32,204	32,204	32,204
Total cost of solar power installation	6346.5	6346.5	6346.5	6346.5
Operating Expenses	38,550	38,550	38,550	38,550

Impacts toward Dilla and BrightWater	
Water coverage to Dilla's population	90
Growth of BrightWater revenue	68.8896624
Growth of BrightWater net earnings	236.7793291
ROI	51.70674138
B/C Ratio	1.567044071
Payback period	1.719078455
CAGR	35.47%

Source: Rainwater Harvesting in Dilla, Ethiopia Feasiblity Study